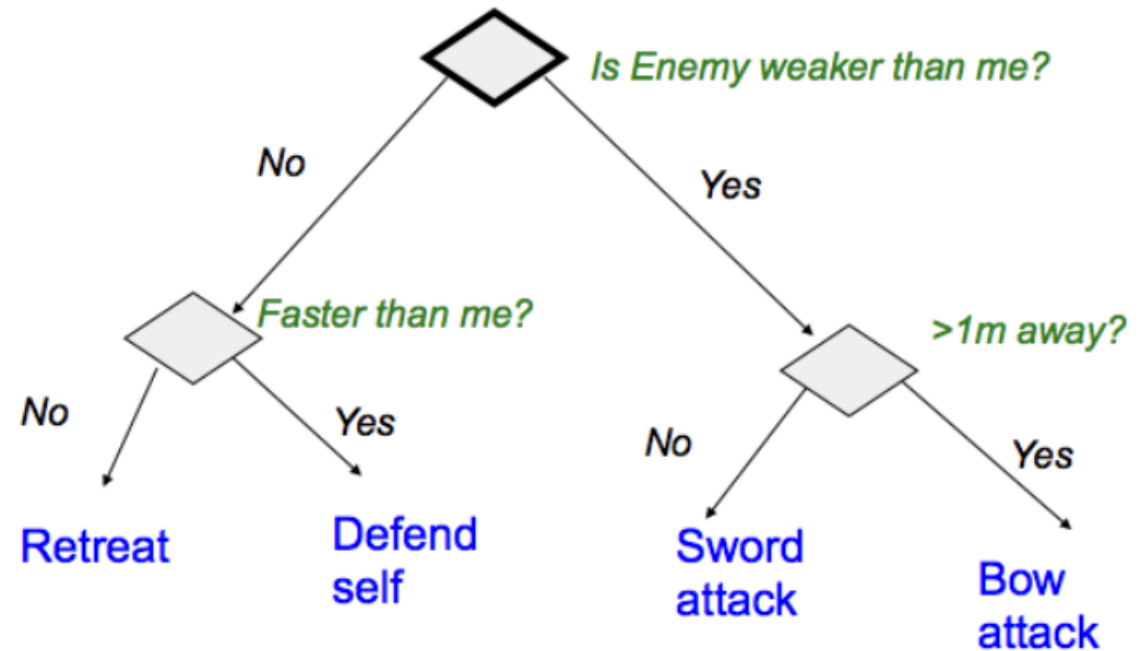


Decizii

- Decizii: miscari, animatii
 - Automate de stare
 - Arbori de decizie
 - Behavior tree
-
- Technicile pot fi aplicate pentru decizii ce trebuie luate la nivel de personaj / prin interactiunea intre personaje.

Decizii

- ◆ **Noduri de decizie:** condiții
- ◆ **Ramuri:** rezultate TRUE/FALSE
- ◆ **Frunze:** acțiuni finale



Utiizare arbori de decizie: obtinere reguli

if is enemy weaker than me and distance to enemy > 1m then Bow attack

ID3?

- Algoritm matematic pentru constructia arborilor de decizie.
- Creat de J. Ross Quinlan in 1979.
- Se bazeaza pe teoria informatiei (Shannon in 1948).
- Construiește arborele de la radacina la frunze.
- Castigul informational este utilizat pentru a selecta atributul pe baza caruia sa se construiasca copiii unui nod.

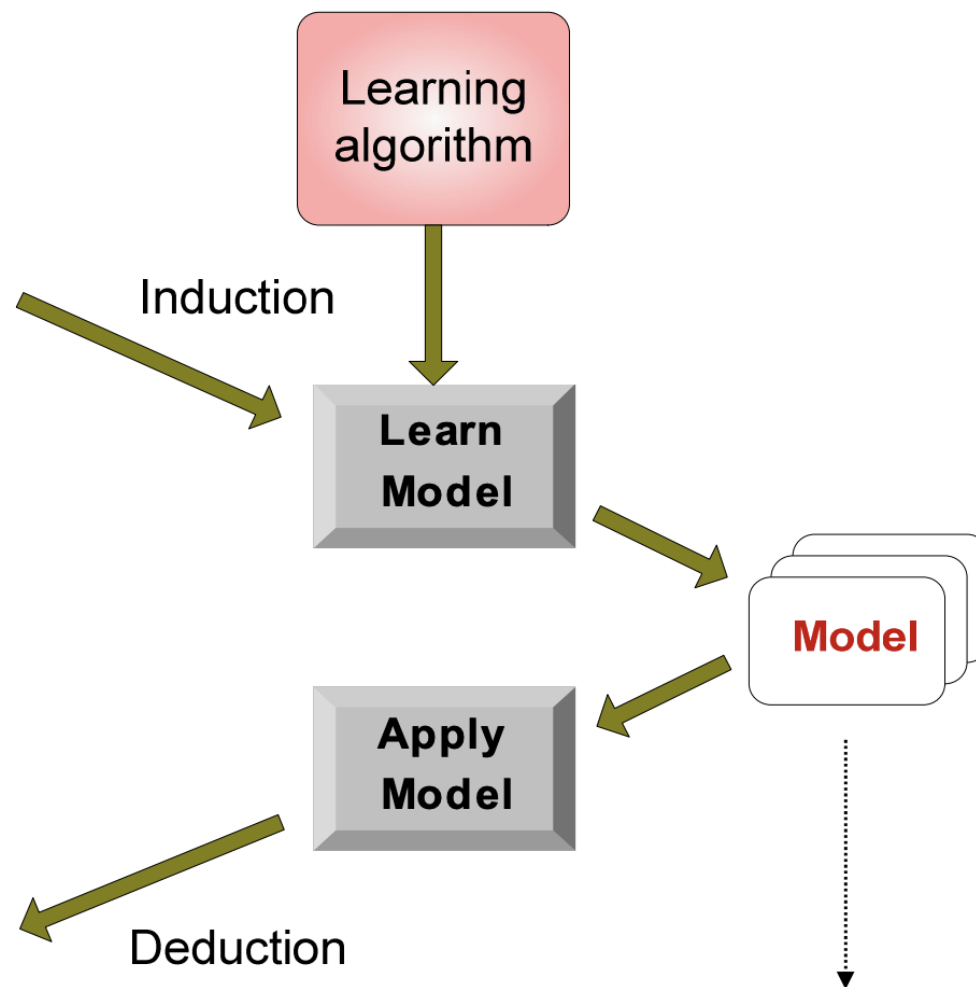
Arbori de decizie

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Set antrenare

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Set testare



Attrib1 = yes $\rightarrow$ Class = No

Attrib1 = No $\wedge$ Attrib3 < 95K $\rightarrow$ Class = Yes

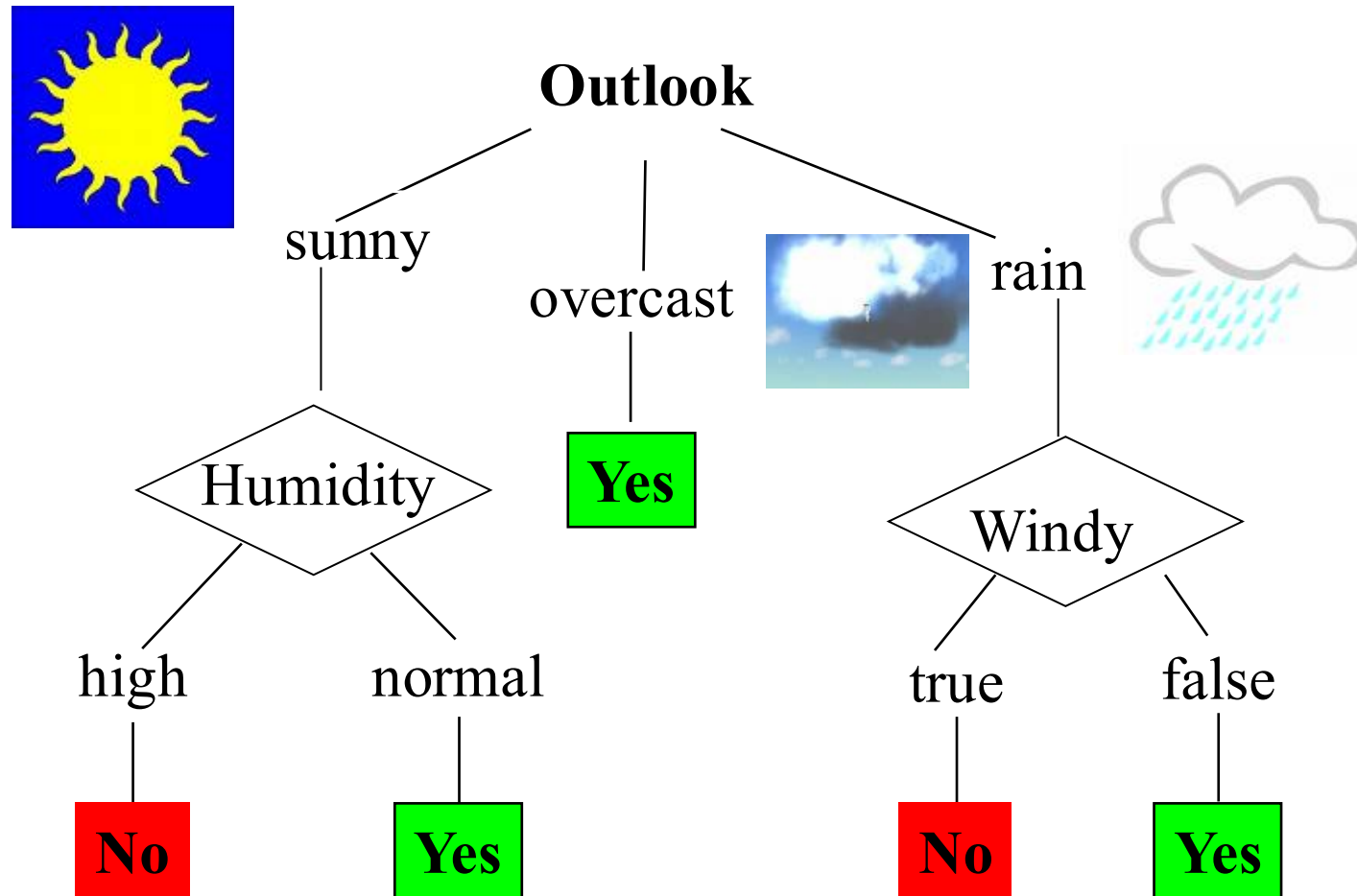
Exemplu arbori de decizie – Play Golf

<i>Day</i>	<i>Temperature</i>	<i>Outlook</i>	<i>Humidity</i>	<i>Windy</i>	<i>Play Golf?</i>
07-05	hot	sunny	high	false	no
07-06	hot	sunny	high	true	no
07-07	hot	overcast	high	false	yes
07-09	cool	rain	normal	false	yes
07-10	cool	overcast	normal	true	yes
07-12	mild	sunny	high	false	no
07-14	cool	sunny	normal	false	yes
07-15	mild	rain	normal	false	yes
07-20	mild	sunny	normal	true	yes
07-21	mild	overcast	high	true	yes
07-22	hot	overcast	normal	false	yes
07-23	mild	rain	high	true	no
07-26	cool	rain	normal	true	no
07-30	mild	rain	high	false	yes

today	cool	sunny	normal	false	?
-------	------	-------	--------	-------	---

tomorrow	mild	sunny	normal	false	?
----------	------	-------	--------	-------	---

Exemplu arbori de decizie – Play Golf



Today: Temperature-cool, Outlook-overcast, Humidity-normal, Windy-false

Tomorrow: Temperature-mild, Outlook-sunny, Humidity-normal, Windy-false

Constructie arbori de decizie – Algoritmul ID3

Entropie

- 2 clase: p_+ (clasa pozitiva) si p_- (clasa negativa)

$$\text{Entropy}(S) = - p_+ \log_2 (p_+) - p_- \log_2 (p_-)$$

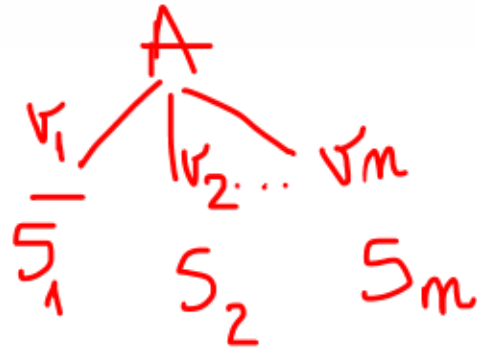
- Formula generala pentru c clase:

$$\text{Entropy}(S) = - \sum_{i=1}^c (p_i \log_2 p_i)$$

Constructie arbori de decizie – Algoritmul ID3

- $\text{Gain}(S, A)$: castigul obtinut pe baza divizarii setului S in functie de atributul A

$$\text{Gain}(S, A) \equiv \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$



Arbori de decizie

- **Outlook = sunny:** 3 exemple yes, 2 exemple no

$$E(\text{Outlook}=\text{sunny}) = -\frac{2}{5} \log\left(\frac{2}{5}\right) - \frac{3}{5} \log\left(\frac{3}{5}\right) = 0.971$$

- **Outlook = overcast:** 4 exemple yes, 0 exemple no

$$E(\text{Outlook}=\text{overcast}) = -1 \log(1) - 0 \log(0) = 0$$

Nedefinit
Valoarea
este 0

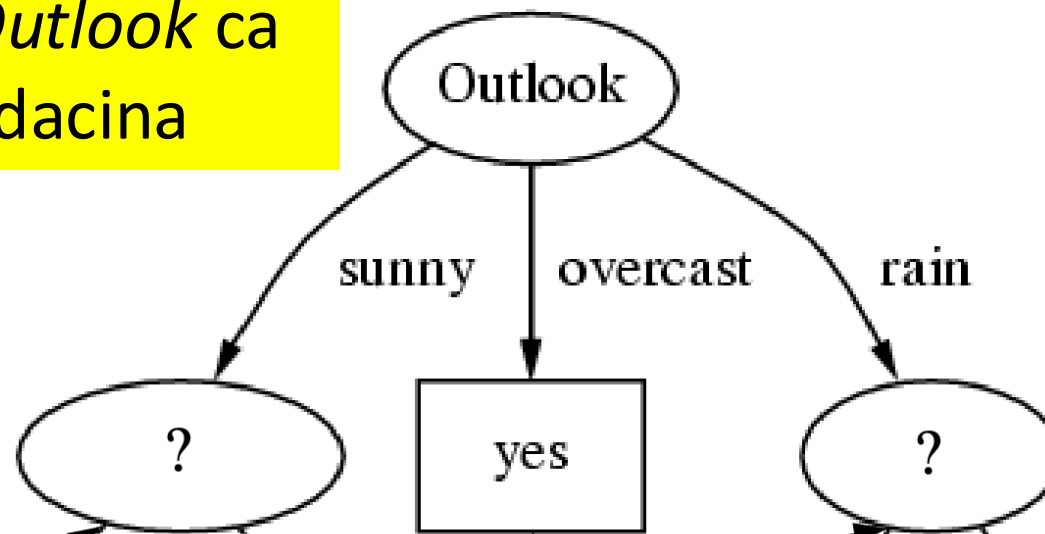
- **Outlook = rain** 2 exemple yes, 3 exemple no

$$E(\text{Outlook}=\text{rain}) = -\frac{3}{5} \log\left(\frac{3}{5}\right) - \frac{2}{5} \log\left(\frac{2}{5}\right) = 0.971$$

$$\text{Gain}(\text{Outlook}) = 0.693$$

Arbori de decizie

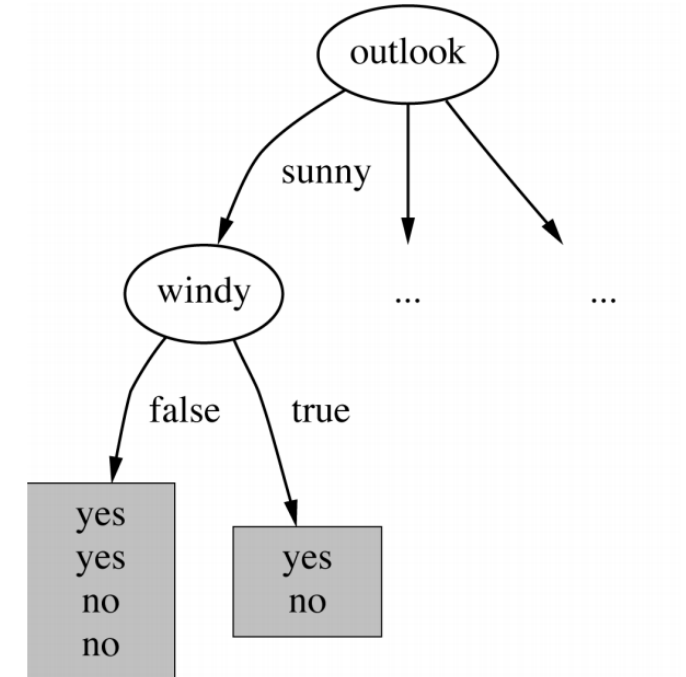
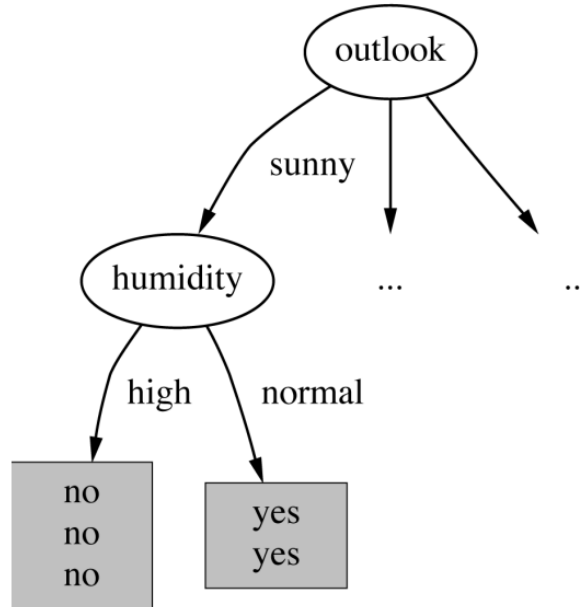
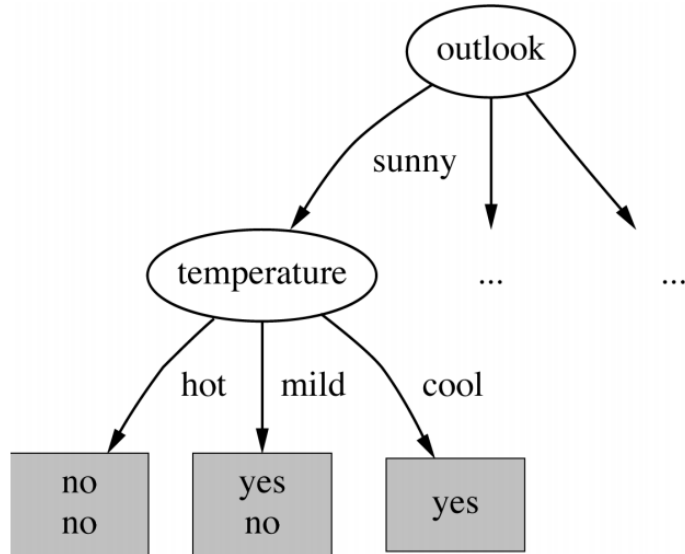
Selectie *Outlook* ca
nod radacina



*Este necesara o noua
divizare*

*Outlook = overcast
contine numai
exemple din clasa yes*

Arbori de decizie



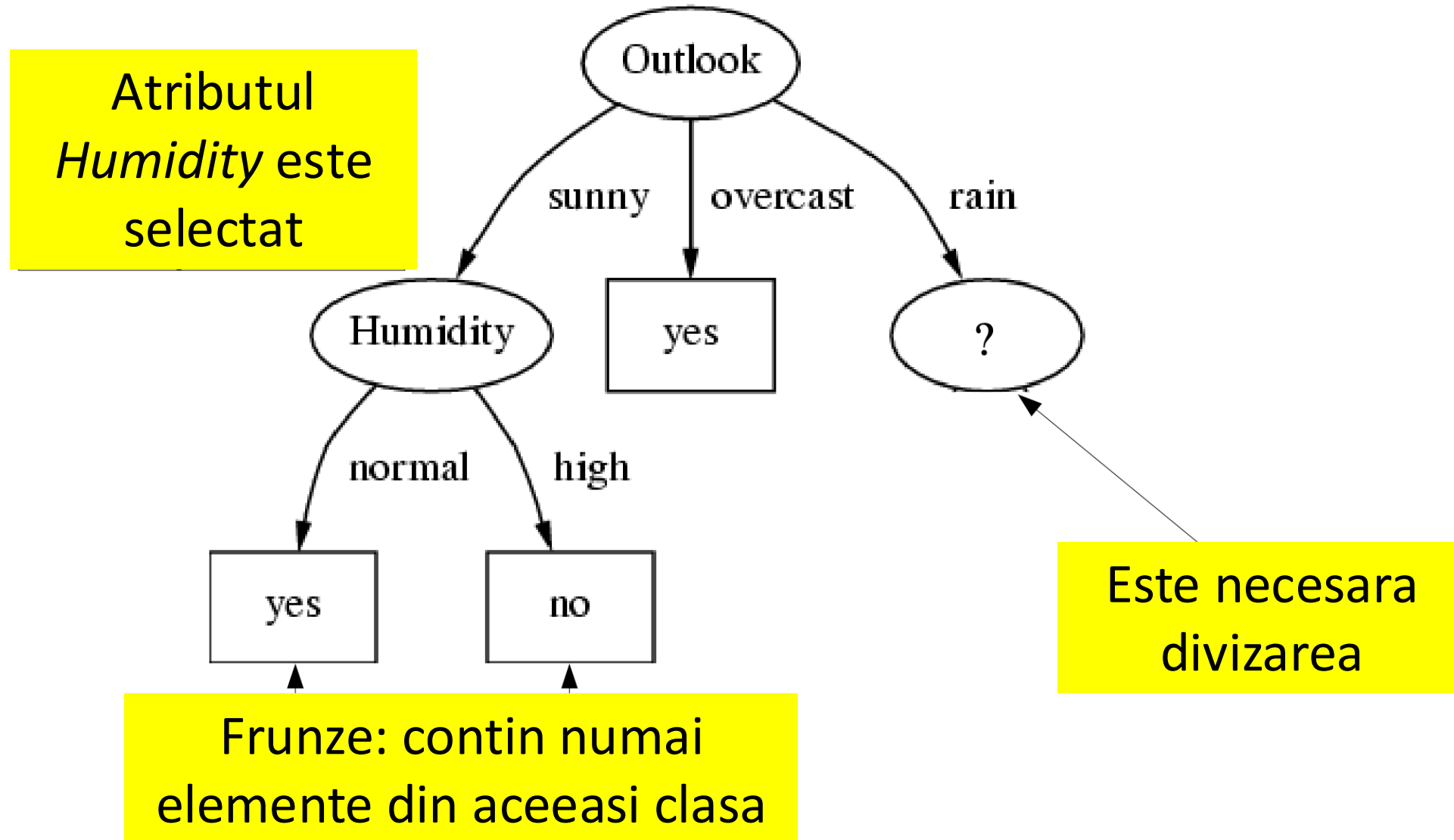
$\text{Gain}(\text{Temperature}) = 0.571$

$\text{Gain}(\text{Humidity}) = 0.971$

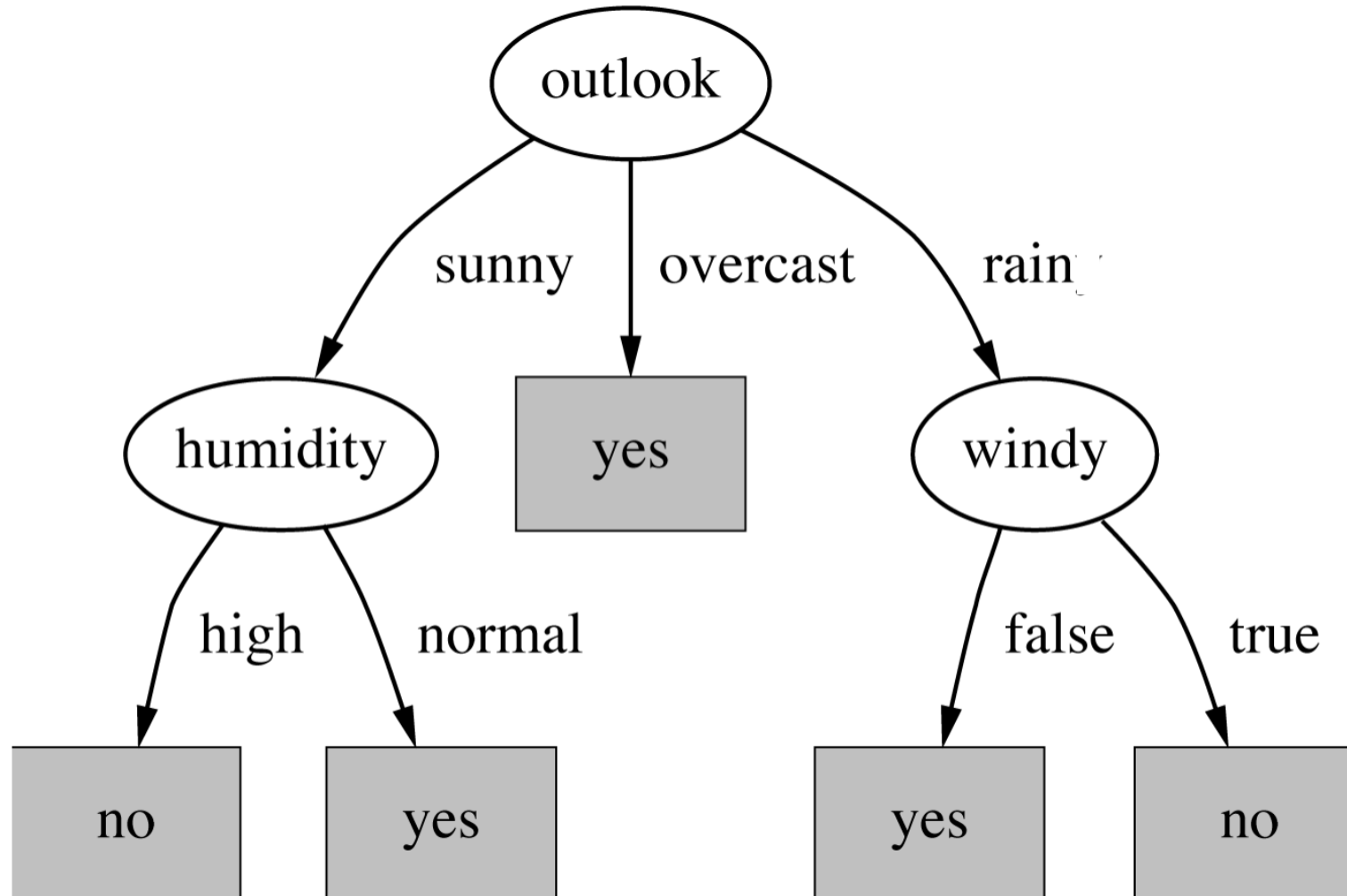
$\text{Gain}(\text{Windy}) = 0.020$

Atributul *Humidity* este selectat

Arbori de decizie



Arbori de decizie



Person	Hair Length	Weight	Age	Class
 Homer	0"	250	36	M
 Marge	10"	150	34	F
 Bart	2"	90	10	M
 Lisa	6"	78	8	F
 Maggie	4"	20	1	F
 Abe	1"	170	70	M
 Selma	8"	160	41	F
 Otto	10"	180	38	M
 Krusty	6"	200	45	M

	Comic	8"	290	38	?
---	-------	----	-----	----	----------

Arbori de decizie – valori numerice

Alegerea pragului dupa care se face divizarea:

1. se sorteaza valorile
2. Se face divizarea intre fiecare pereche de valori

64	65	68	69	70	71	72	72	75	75	80	81	83	85
Yes	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	Yes	Yes	No

- One split between each pair of values

Temperature < 71.5 : yes/4, no/2

Temperature ≥ 71.5 : yes/5, no/3

Divizarea se poate face alegand valoarea medie sau valoarea curenta

Arbori de decizie – valori numerice

Alegerea pragului dupa care se face divizarea:

1. se sorteaza valorile
2. Se identifica exemplele care difera in clasificare

<i>Temperature:</i>	40	48	60	72	80	90
<i>EnjoyTennis:</i>	No	No	Yes	Yes	Yes	No

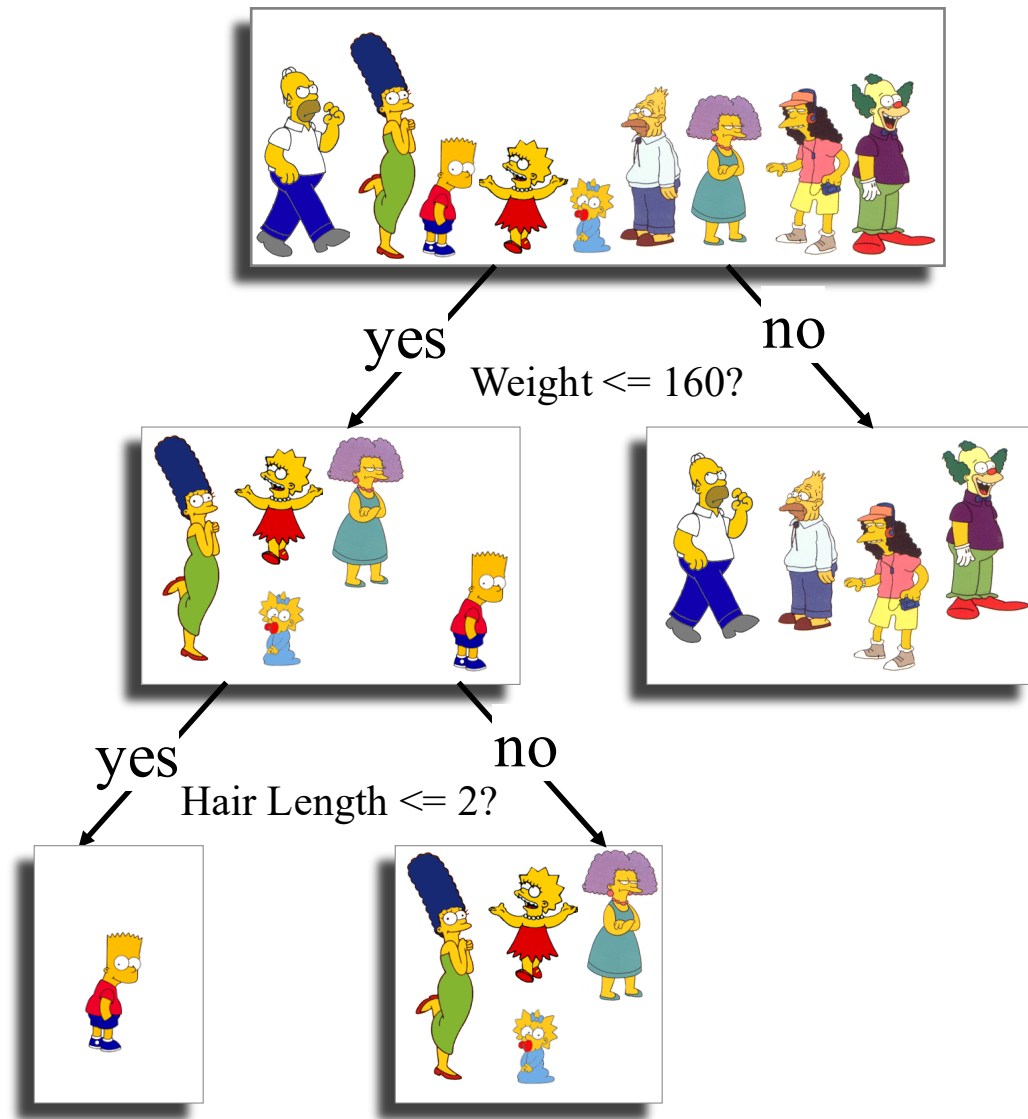
*Temperature*_{>54}

*Temperature*_{>85}

Atributul *Weight* are gain-ul cel mai bun.

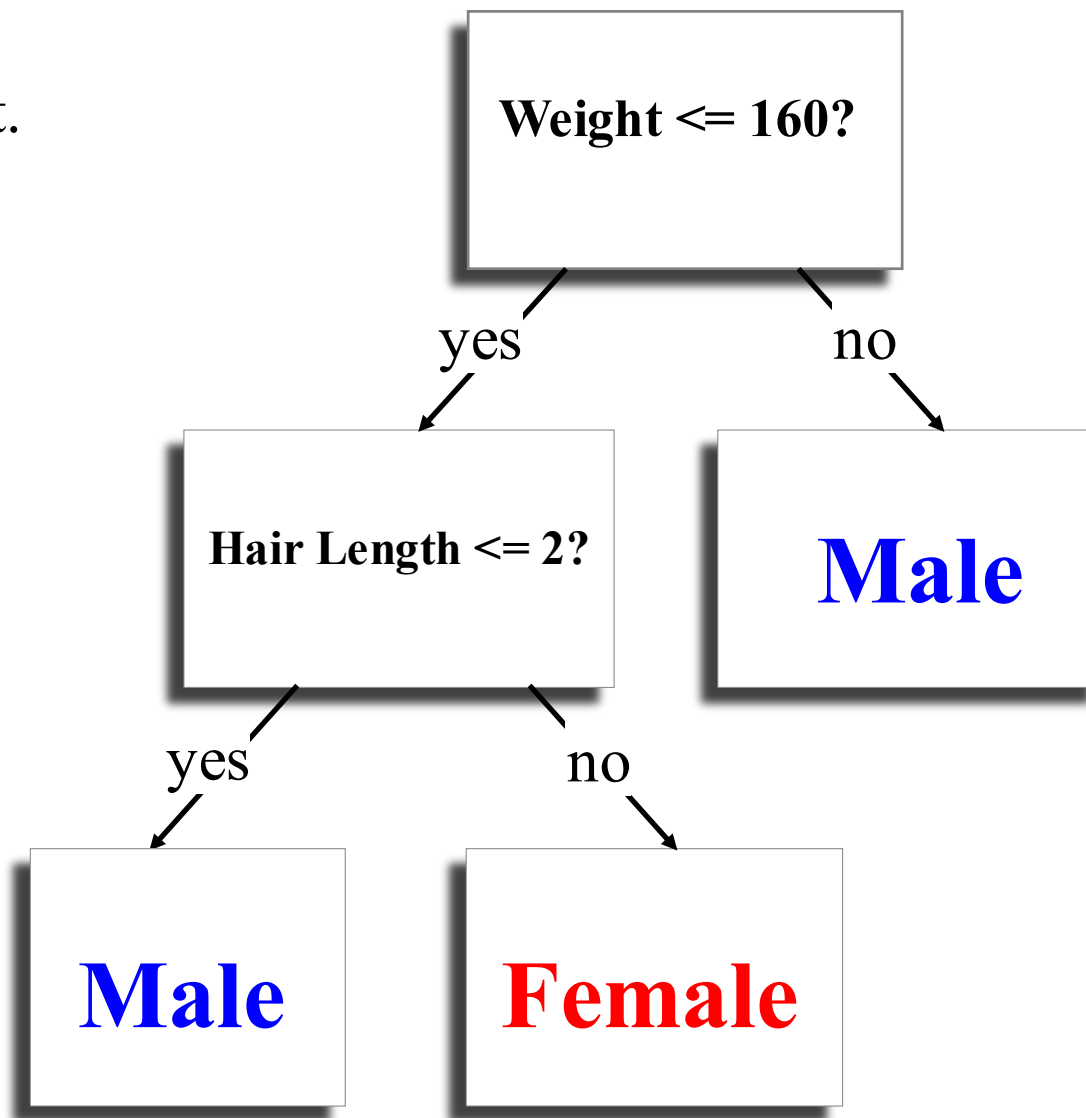
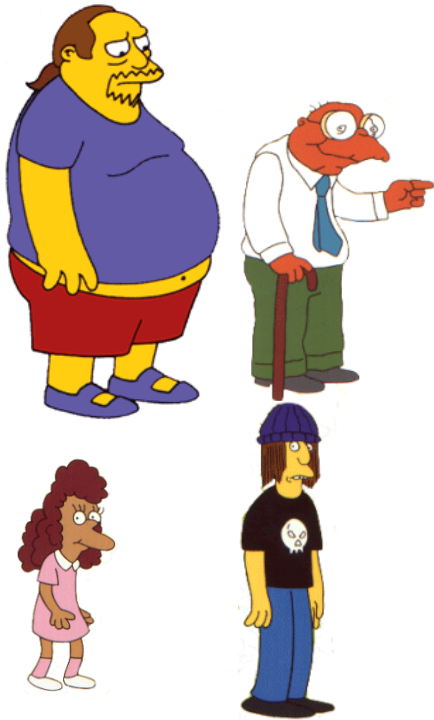
Persoane cu $Weight > 160$ sunt casificate correct.

Pentru restul persoanelor se continua clasificarea, folosind restul atributelor → atributul cu gain-ul cel mai bun este *Hair length*.

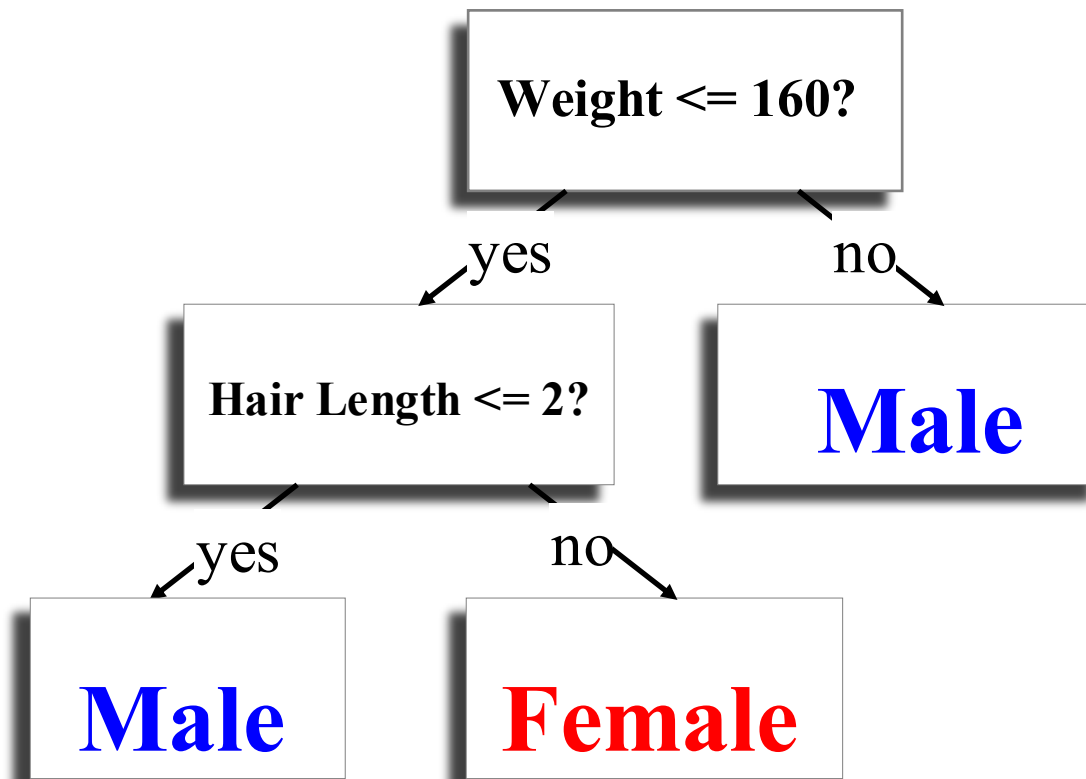


Arborele pastreaza conditiile de test.

Cum se clasifica
aceste date?



Conversie arbori de decizie -> reguli



Reguli pentru clasificare to barbati/femei

If *Weight* greater than 160, classify as **Male**

Else if *Hair Length* less than or equal to 2, classify as **Male**

Else classify as **Female**

Avantaje

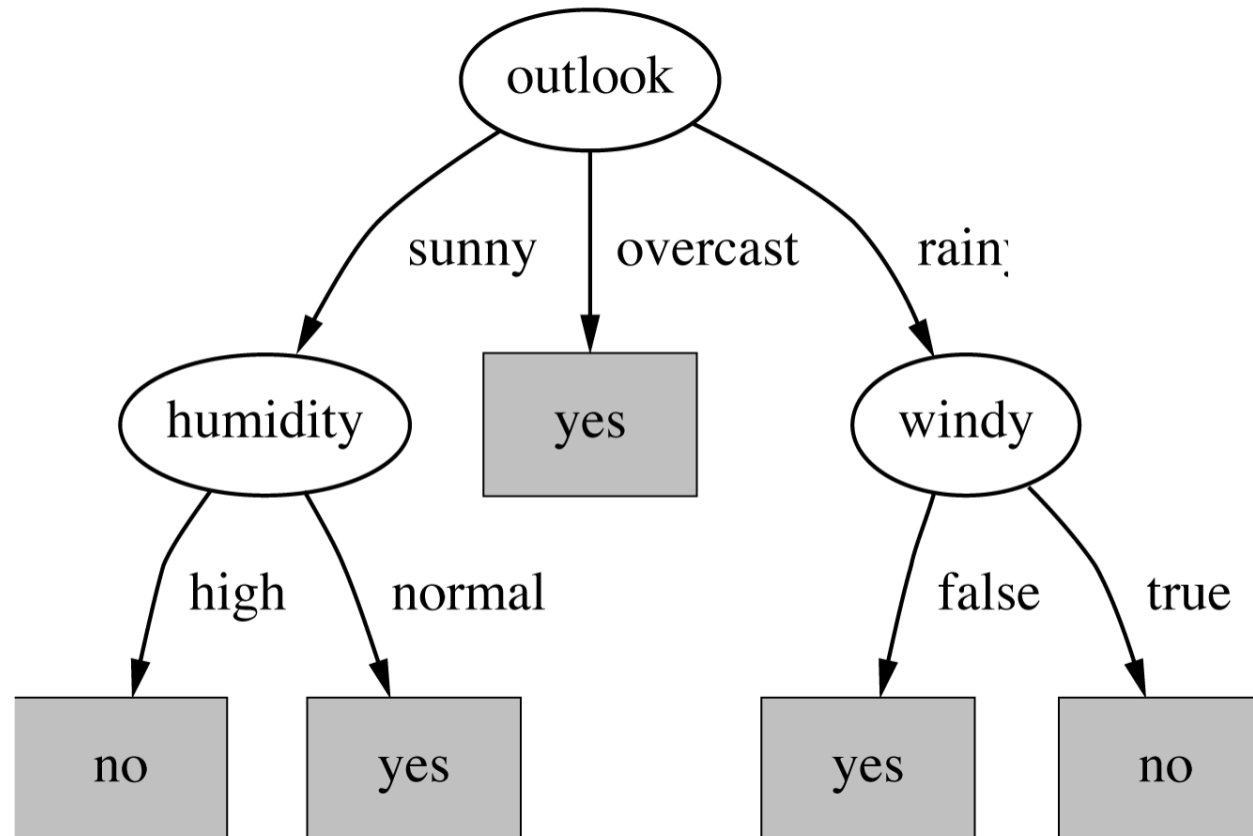
- **Simplitate:** usor de inteles si implementat.
- **Performanță:** eficient (evaluari de conditii).
- **Explicabilitate:** se poate vedea ramura selectata.

Dezavantaje

- **Scalabilitate slaba:** adaugarea de conditii face ca arborele sa devina foarte mare -> dificil pentru AI complex
- **Repetitie de logica:** ramuri similare pot fi duplicate.
- **Nu gestioneaza bine actiuni de durata**

Probleme

Input: (?, sunny, normal, false, **no**) → **Yes**

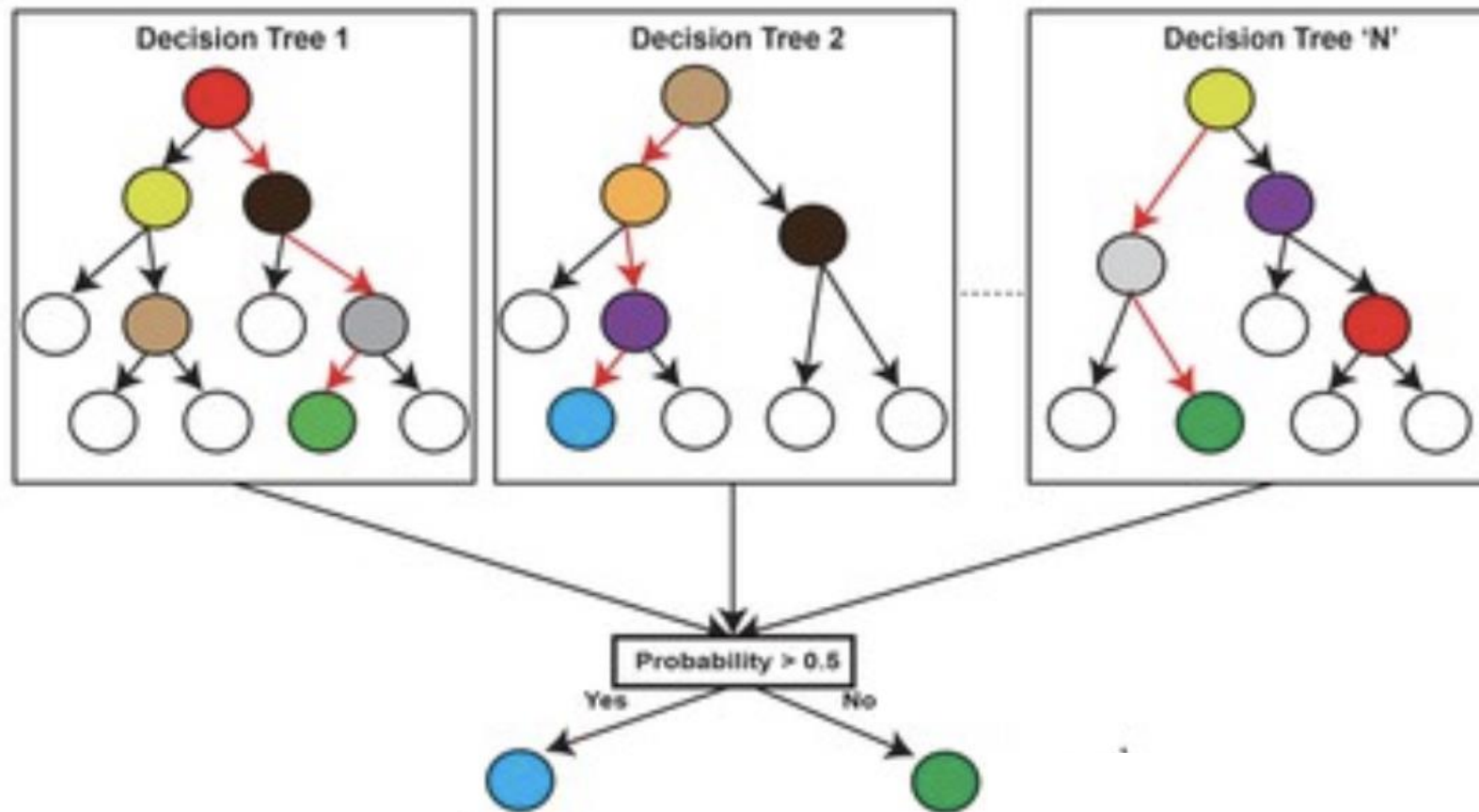


Probleme

- Modelul este prea simplu
- Seturile de date de antrenare si testare au erori mari
- Valori lipsa:
Input: (?, sunny, normal, false, no)

Random Forest

- N random trees
- Votarea clasei majoritare
- Mai robust decat un arbore de decizie



Arbori de comportament (Behavior tree)

- Organizare in noduri compuse modular care sunt evaluate periodic
 - **Radacina**
 - **Noduri compuse** (Selector, Sequence)
 - **Frunze** (actiuni)
- Inamic:
 - Daca vede jucatorul → ataca
 - Daca nu vede jucatorul, dar aude un zgomot → investigheaza
 - Daca nu vede si nu aude nimic → patruleaza

```
Selector (Root)
|
├─ Sequence (Atac)
|   └─ Condiție: Vede jucătorul?
|       └─ Acțiune: Atacă
|
├─ Sequence (Investigare)
|   └─ Condiție: Aude zgomot?
|       └─ Acțiune: Mergi la sursa zgomotului
|
└─ Acțiune: Patrulează
```

Arbori de comportament (Behavior tree)

- **Nod Selector** - incearca fiecare copil de sus in jos - se opreste la primul care are success
- **Nod Sequence** - executa nodurile in ordine - daca unul esueaza → toata secventa esueaza
- Noduri frunza: intorc SUCCESS sau FAILURE

```
Selector (Root)
|
├─ Sequence (Atac)
|   └─ Condiție: Vede jucătorul?
|       └─ Acțiune: Atacă
|
├─ Sequence (Investigare)
|   └─ Condiție: Aude zgomot?
|       └─ Acțiune: Mergi la sursa zgomotului
|
└─ Acțiune: Patrulează
```

Arbori de comportament (Behavior tree)

- Selectorul:
- Vede jucatorul?
 - Da $\rightarrow$ Ataca $\rightarrow$ STOP
 - Nu $\rightarrow$ merge la urmatoarea ramura
- Aude zgomot?
 - Da $\rightarrow$ Investigare $\rightarrow$ STOP
 - Nu $\rightarrow$ Patruleaza

```
Selector (Root)
|
├─ Sequence (Atac)
|   └─ Condiție: Vede jucătorul?
|       └─ Acțiune: Atacă
|
├─ Sequence (Investigare)
|   └─ Condiție: Aude zgomot?
|       └─ Acțiune: Mergi la sursa zgomotului
|
└─ Acțiune: Patrulează
```

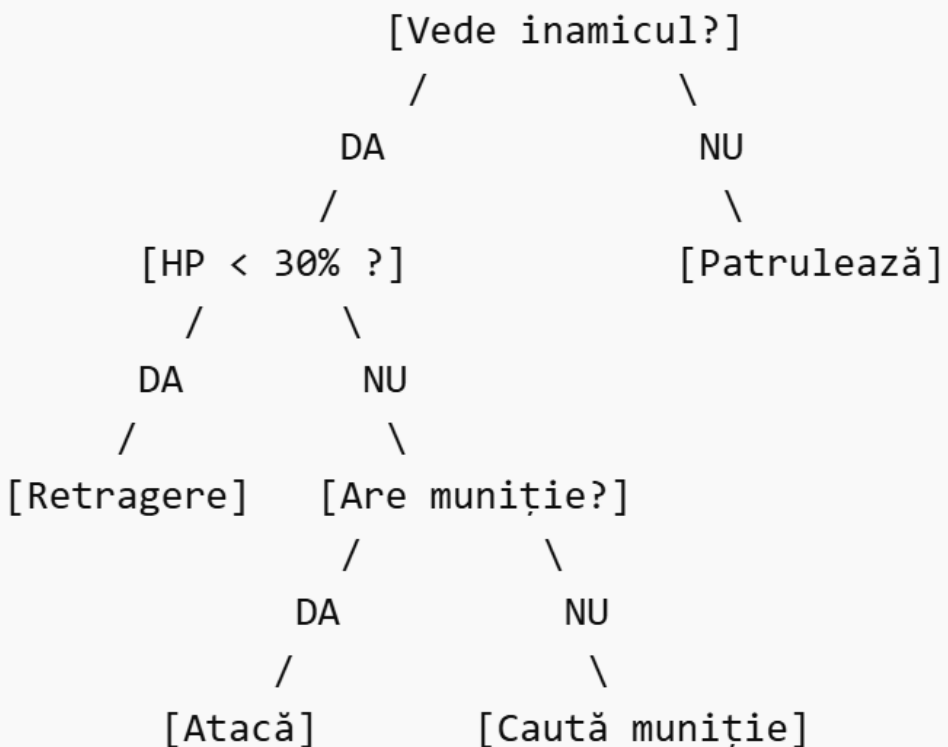
Arbori de comportament (Behavior tree)

- Modular si reutilizabil: Modificarea si extinderea cu usurinta a comportamentelor
 - Scalabil: Functioneaza bine pentru AI simple si complexe
 - Structura mai flexibila decat automatele de stare
-
- Arbori de decizie = Care este decizia finala in aceasta situatie?
 - Behavior Tree = Care este comportamentul meu continuu?

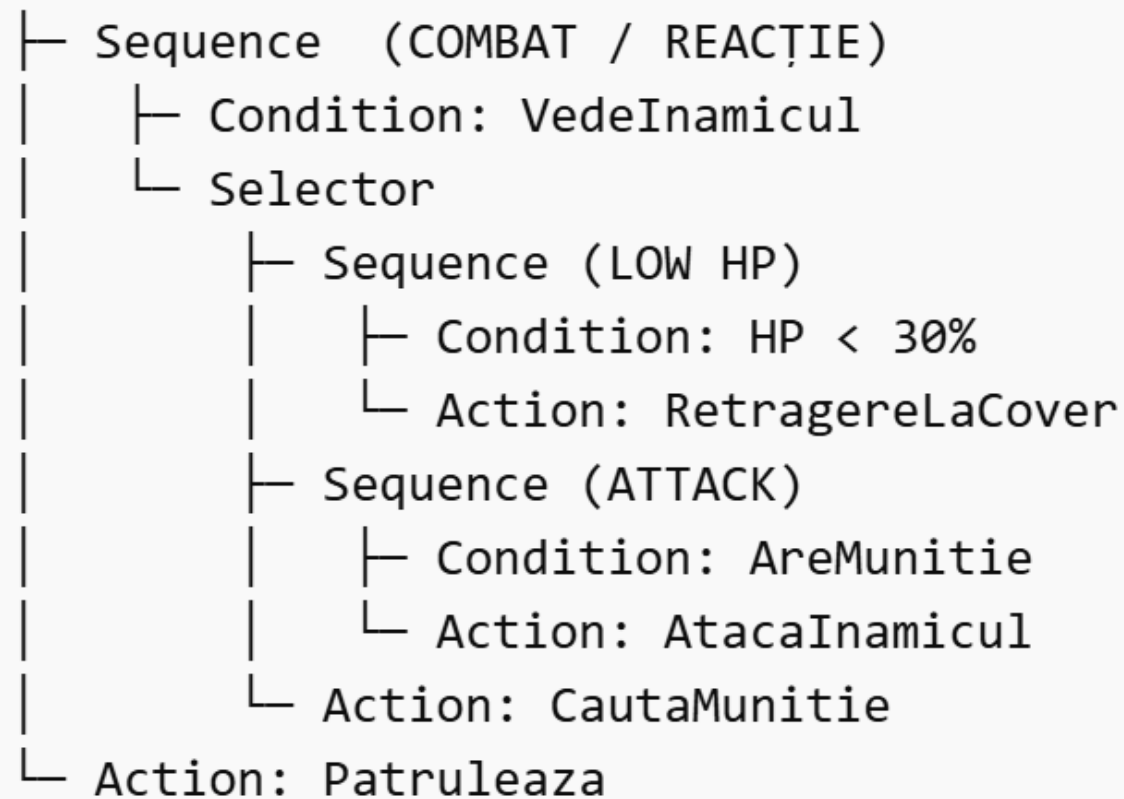
Daca vede inamicul:

- daca are viata mica → se retrage
- daca are munitie → ataca
- daca nu are munitie → cauta munitie

•Daca nu vede inamicul → patruleaza



ROOT: Selector



Combinare arbori de decizie – arbori de comportament

- Luare mai bună a deciziilor – arborele de decizie alege cea mai bună strategie, în timp ce arborele de comportament asigură o execuție fluidă.
- Scalabilitate: Arbori de decizie: se ocupă de logică, în timp ce arborii de comportament gestionează acțiunile reutilizabile.
- Inteligență artificială mai realistă: Inamicii pot lua decizii mai inteligente pe baza condițiilor.
- Eficientă și modularitate: Păstrează codul AI organizat și reutilizabil.

Abordare hibrida

arbori de decizie – arbori de comportament

- Jucatorul se apropie de un inamic
 - Health: 60%, Distance: 3m**

1: Decision Tree – selecteaza o strategie

- Health ~ moderate si jucator apropiat** se selecteaza "**ATTACK**".

2: Behavior Tree – selecteaza cea mai buna actiune

- Jucator foarte apropiat de inamic**, se selecteaza "**Melee Attack**" in loc de "**Ranged Attack**".

Abordare hibrida

arbori de decizie – arbori de comportament

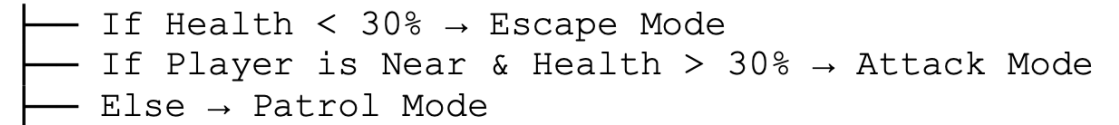
1. Decision Tree - High-Level Choices

- Evaluateaza conditiile din joc si alege strategia
- Exemplu:
 - *Is the enemy stronger than the player?* → Flee
 - *Is the player nearby?* → Attack
 - *Is the player far?* → Search for the player

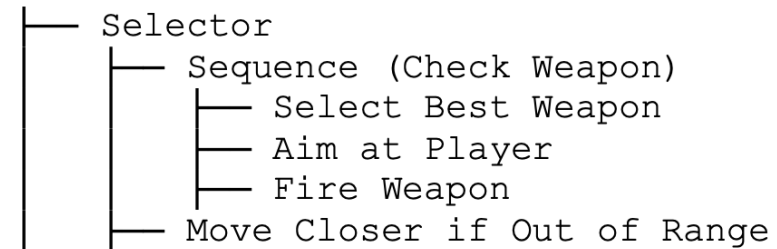
2. Behavior Tree – se executa actiunile pe baza selectiei facute de Decision Tree

- The **Behavior Tree** separa strategia aleasa in actiuni componente.
- Exemplu:
 - *Attack Strategy* → Move to player → Choose best weapon → Attack
 - *Flee Strategy* → Find nearest cover → Move to cover → Hide

Decision Tree:

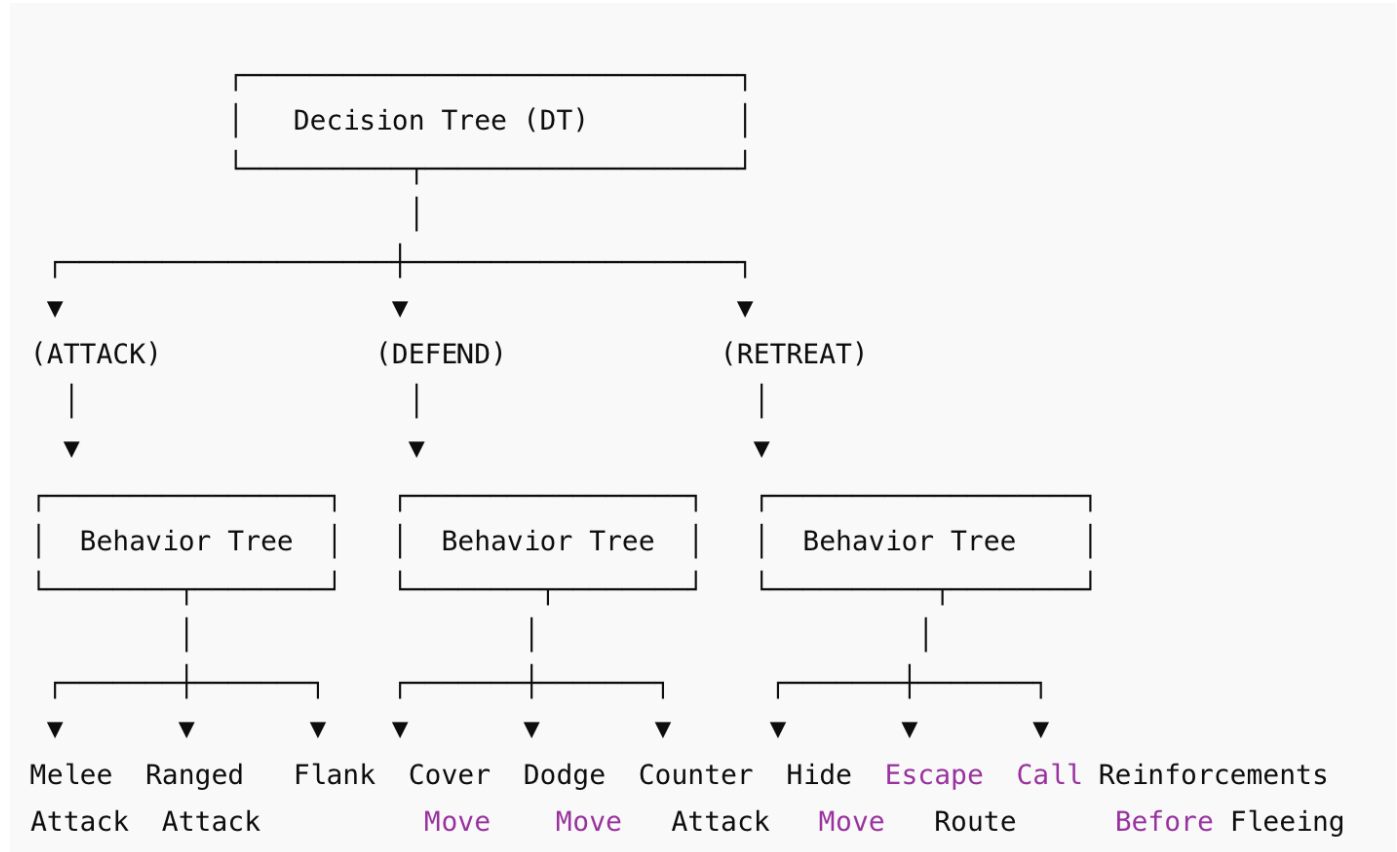


Behavior Tree for Attack Mode:



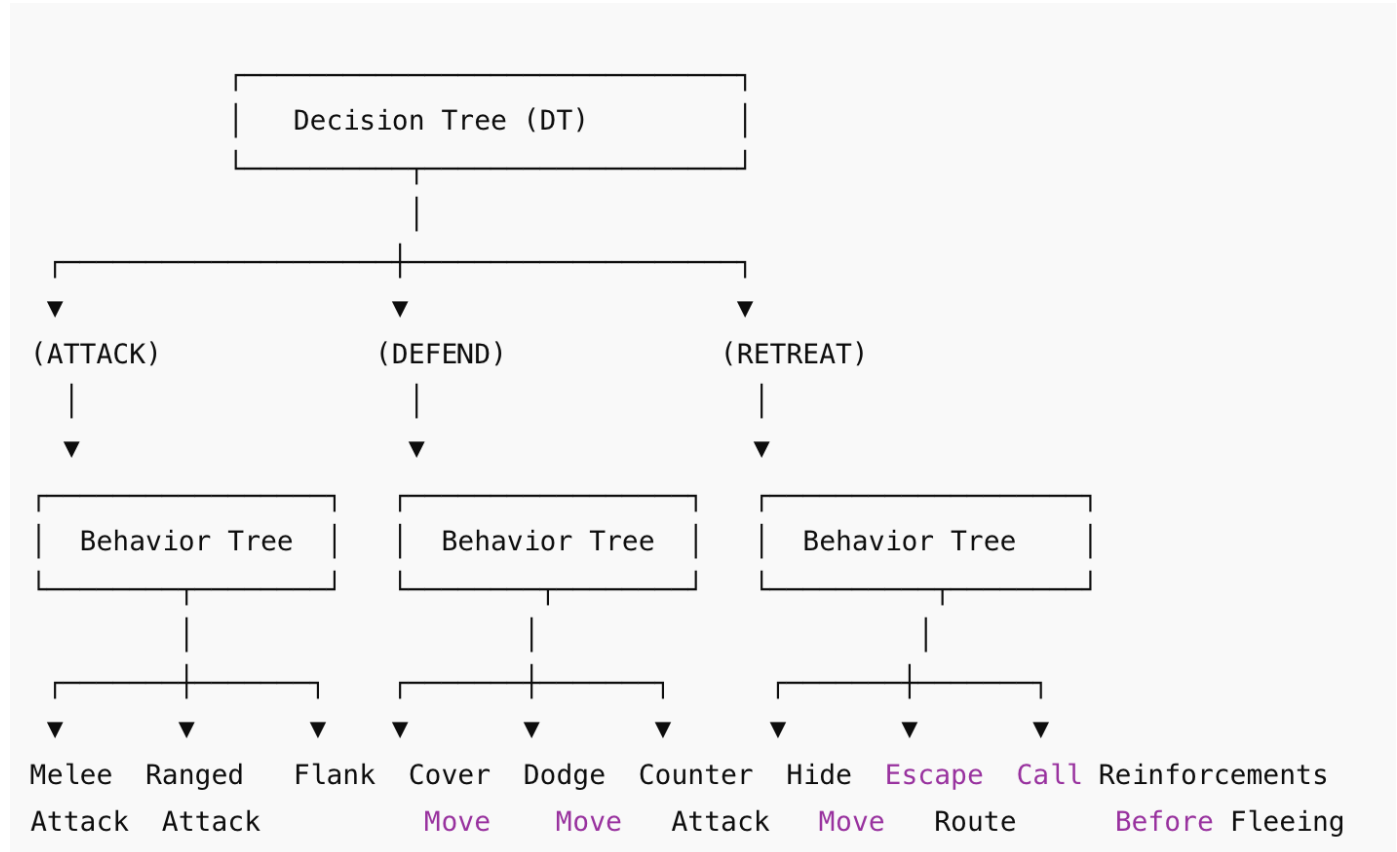
Abordare hibrida

- **Player visibility** (Hidden, Spotted, Engaged in Combat)
- **Enemy health** (Full, Injured, Critical)
- **Backup availability** (Alone, Nearby Allies, Reinforcements Coming)



Abordare hibrida

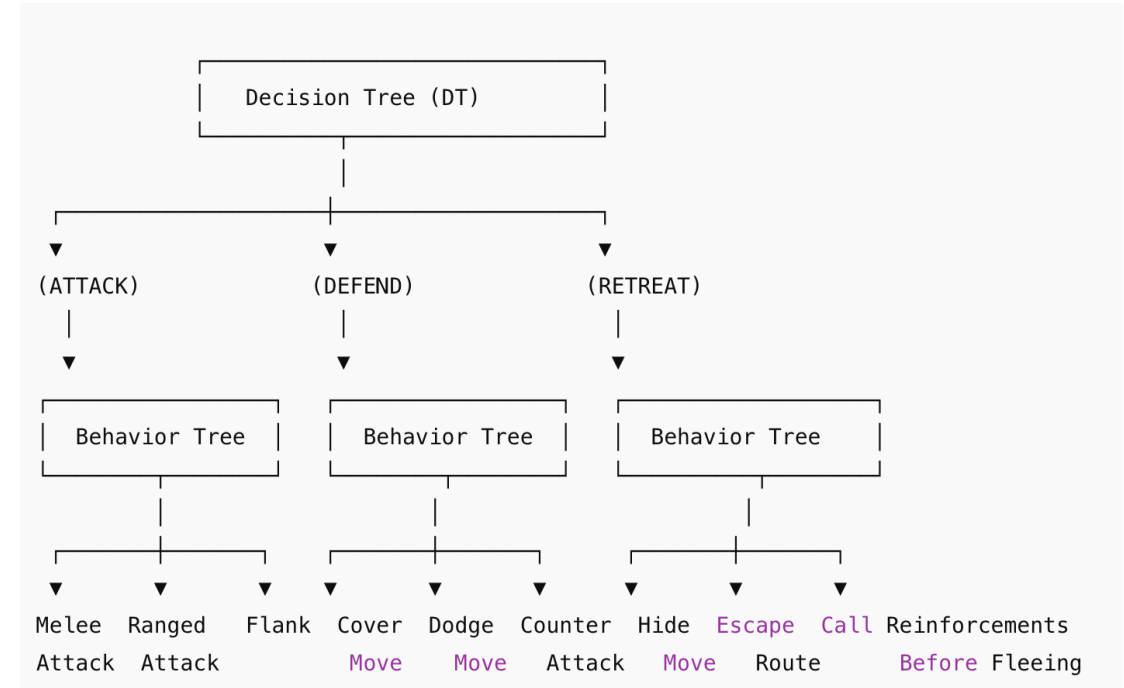
- Decision Tree selecteaza modalitatea de abordare
- **Attack** (If the enemy is healthy and has backup)
- **Defend** (If the enemy is injured but has cover available)
- **Retreat** (If critically injured and alone)



Abordare hibrida

Behavior Tree rafinare strategie:

- **Attack Strategy**
 - **Check Weapon Type**
 - If Melee → Charge at player
 - If Ranged → Take cover and fire
 - **Coordinate with Allies**
 - If reinforcements available → Flank the player
- **Defend Strategy**
 - **Find Cover**
 - If cover available → Move to cover
 - If no cover → Dodge and block attacks
 - **Counterattack if Safe**
 - If the player reloads → Attack
- **Retreat Strategy**
 - **Find Exit Route**
 - Move toward the nearest escape point
 - **Use Stealth**
 - Hide behind obstacles
 - Call for reinforcements before fleeing



Abordare hibrida

	Arbori de decizie	Arbori de comportament	Hibrid
Decizii	Rapid, rigid	Flexibil	Balansat
Adaptabilitate	Greu de modificat	Modular & adaptiv	Adaptivitate ridicata
Scalabilitate	Complex	Usor de extins	Cel mai bun
Viteza executiei	Decizii rapide	Mai lent, dar structurat	Optimizat